SIAM/ACM Proceedings Macros for Use With LaTeX*

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Abstract. The abstract must be able to stand alone and so cannot contain citations to the paper's references, equations, etc. An abstract must consist of a single paragraph and be concise. Because of online formatting, abstracts must appear as plain as possible. Any equations should be inline.

1 Introduction. The introduction introduces the context and summarizes the manuscript. It is important to clearly state the contributions of this piece of work.

2 Cross references and hyperlinks. The SIAM Proceedings macros support cross references and hyperlinks via the `cleveref` and `hyperef` packages, which are loaded by default.

2.1 Cleveref. SIAM strongly recommends using the commands provided by the `cleveref` package for cross referencing. The package is automatically loaded and already customized to adhere to SIAM's style guidelines. To create a cross reference, use the command `\cref` (inside sentence) or `\Cref` (beginning of a sentence) in place of the object name and `\ref`. The `cleveref` package enhances LaTeX's cross-referencing features, allowing the format of cross references to be determined automatically according to the "type" of cross reference (equation, section, etc.) and the context in which the cross reference is used. So, the package *automatically* inserts the object name as well as the appropriate hyperlink. Additional examples are shown in the following sections for equations, tables, figures, sections, etc.

2.2 Hyperef. Hyperlinks are created with the `\href` and `\url` commands. The `\email` command is also defined, as shown in the Page 1 footnote below.

3 Math and equations. Here we show some example equations, with numbering, and examples of referencing the equations. The SIAM Proceedings macro includes the package `amsmath` by default, and we include some of its features as well, although the reader should consult the package user manual for further guidance [2, 5]. Note that several of the example are adapted from Mittlebach and Goossen's guide to LaTeX [7].

Here is a straightforward example of inline mathematics equations that does not use any special packages or features: Let $S = [s_{ij}]$ $(1 \leq i, j \leq n)$ be a $(0, 1, -1)$-matrix of order n.

Next we show the `smallmatrix` environment for an inline matrix from the `amsmath` package, which is included by default.

Matrices of no more than two rows appearing in text can be created as shown: $B = \left[\begin{smallmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{smallmatrix}\right]$.

Bigger matrices can be rendered with environments from the `amsmath` package, such as `bmatrix` and `pmatrix`:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{3.1}$$

Equation (3.1) above shows some example matrices.

The following shows how to use the `align` environment from `amsmath` to easily align multiple equations.

Equations (3.2)–(3.4) show three aligned equations.

$$\begin{align} f &= g, \tag{3.2} \\ f' &= g', \quad \text{and} \tag{3.3} \\ \mathcal{L}f &= \mathcal{L}g. \tag{3.4} \end{align}$$

Another way to number a set of equations is the `subequations` environment from `amsmath`.

We calculate the Fréchet derivative of F as follows:

$$\begin{align} F'(U,V)(H,K) =& \langle R(U,V), H\Sigma V^T \tag{3.5a} \\ & + U\Sigma K^T - P(H\Sigma V^T \notag \\ & + U\Sigma K^T)\rangle \notag \\ =& \langle R(U,V), H\Sigma V^T + U\Sigma K^T \rangle \notag \\ =& \langle R(U,V)V\Sigma^T, H\rangle \tag{3.5b} \\ & + \langle \Sigma^T U^T R(U,V), K^T\rangle. \notag \end{align}$$

We can link to Equation (3.5a) at the beginning of the equation and (3.5b) later in the equation separately.

*The full version of the paper can be accessed at https://arxiv.org/abs/0000.00000

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For an equation split over multiple lines, the next equation shows the usage of the `multline` environment provided by `amsmath`.

We claim that the projection $g(U, V)$ is given by the pair of matrices:

$$g(U,V) = \left(\frac{R(U)V\Sigma^T U^T - U\Sigma V^T R(U)^T}{2} U, \right. \\ \left. \frac{R(U)^T U\Sigma V^T - V\Sigma^T U^T R(U)}{2} V \right). \tag{3.6}$$

4 Theorem-like environments. The SIAM Proceedings macro loads the `ntheorem` package and uses it to define the following theorem-like environments: `theorem`, `lemma`, `corollary`, `definition`, and `proposition`. SIAM also defines a `proof` environment that automatically inserts the symbol "□" at the end of any proof, even if it ends in an equation environment. *Note that the document may need to be compiled twice for the mark to appear.* Some of the calculus examples were adapted from [4]. The following shows usage of the `theorem` environment. Here we state our main result as Theorem 4.1.

THEOREM 4.1 (LDL^T Factorization [6]). *If $A \in \mathbb{R}^{n\times n}$ is symmetric and the principal submatrix $A(1:k, 1:k)$ is nonsingular for $k = 1:n-1$, then there exists a unit lower triangular matrix L and a diagonal matrix*

$$D = \operatorname{diag}(d_1, \dots, d_n)$$

such that $A = LDL^T$. The factorization is unique.

Followed by a theorem without an optional argument used to name the theorem.

THEOREM 4.2. *Suppose f is a function that is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b). Then there exists a number c such that $a < c < b$ and*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

In other words,

$$f(b) - f(a) = f'(c)(b - a).$$

Observe that Theorems 4.1 and 4.2 and Corollary 4.3 correctly mix references to multiple labels.

COROLLARY 4.3. *Let $f(x)$ be continuous and differentiable everywhere. If $f(x)$ has at least two roots, then $f'(x)$ must have at least one root.*

Proof. Let a and b be two distinct roots of f. By Theorem 4.2, there exists a number c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{0 - 0}{b - a} = 0.$$ □

Note that we can also alter the proof environment.

Proof of Theorem 4.2. We now show... □

The macro also defines commands to create your own theorem- and remark-like environments:

- `newsiamthm` — Small caps header, italized body.
- `newsiamremark` — Italics header, roman body.

Each command takes two arguments. The first is the environment name, and the second is the name to show in the document. These commands should be used instead of `\newtheorem`.

NOTE: Due to a current limitation of `cleveref`, any custom defined theorem-like environment aside from the ones listed in the SIAM Style Manual [9, Chapter 6] will need to use `\ref` instead of `\cref` when referenced in the document. The theorem-like environments listed in the style manual that are safe to use with `cleveref` are `theorem`, `lemma`, `corollary`, `definition`, `proposition`, `fact`, `claim`, `conclusion`, `result`, `consequence`, and `problem`.

An additional note is that `fact`, `claim`, `conclusion`, `result`, `consequence`, and `problem` can also be defined as a remark-like environment in accordance with the SIAM Style Manual [9, Chapter 6] without having an issue with `cleveref` if an author chooses to do so.

5 Figures. It is recommended that all figures be generated in a vector format (e.g., EPS or PDF). If figures in a JPG or PNG format are used, make sure they are of a sufficiently high resolution. Make sure that your figures are optimized! For instance, EPS and PDF figures should not have thousands of layers that vastly increase the file size for no benefit to image legibility.

The following is an example for how to generate Figure 5.1. This example uses the `graphicx` package for the `\includegraphics` command.

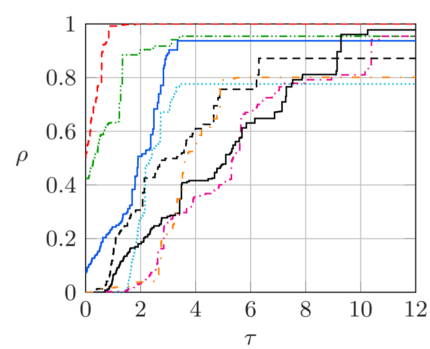

Figure 5.1: Example figure using an external image file.

If working with EPS images and using `pdflatex`, we use the package `epstopdf` to automatically convert EPS images to PDF for inclusion in PDF documents created by `pdflatex`. Please note that `epstopdf` requires `texlive-font-utils` which may not be part of a standard LaTeX installation.

6 Tables. Table captions should go above the tables. The following shows the code to generate Table 6.1. More complicated tables can be created using subfloats via the `subfig` package, as well as special column options from the `array` package.

Table 6.1: Example table

Species	**Mean**	**Std. Dev.**
1	3.4	1.2
2	5.4	0.6
3	7.4	2.4
4	9.4	1.8

7 Algorithm. The `algorithm` package is automatically included in the SIAM Proceedings macro. This provides the float environment. Users have the choice of `algpseudocode`, `algorithmic`, and other packages for actually formatting the algorithm. For example, Algorithm 7.1 is produced as follows.

Algorithm 7.1 Build tree

Define $P := T := \{\{1\}, \ldots, \{d\}\}$
while $\#P > 1$ **do**
  Choose $C' \in \mathcal{C}_p(P) \in C' := \operatorname{argmin}_{C \in \mathcal{C}_p(P)} \varrho(C)$
  Find an optimal partition tree $T_{C'}$
  Update $P := (P \backslash C') \cup \{\bigcup_{t \in C'} t\}$
  Update $T := T \cup \{\bigcup_{t \in \tau} t : \tau \in T_{C'} \backslash \mathcal{L}(T_{C'})\}$
end while
return T

8 Sections. Sections are denoted using standard LaTeX section commands, i.e., `\section`, `\subsection`, etc. Section heading names should end with a period to separate them from the content that follows.

Any numbered, labeled sections can be referenced using `\cref`, including those without a title.

8.1 Lower level sections. Note that there may be some variation between different proceedings regarding the use of lower level sections. While this SIAM Proceedings macro does support the use of `\subsubsection`, `\paragraph`, and `\subparagraph`, authors should follow the guidelines for a given proceedings on what is and is not allowed.

Paragraph. Here is an example of a `\paragraph`. Paragraph and subparagraph headings are italic and not numbered.

Subparagraph. Here is an example of a `\subparagraph`.

8.1.1 Subsubsection. Here is an example of a `\subsubsection`.

9 SIAM style considerations. When not specifically mentioned in the given guidelines for a specific proceedings, SIAM proceedings articles should follow SIAM's editorial style manual [9]. For instance, the serial/Oxford comma is set up in `cleveref` by default and expected to be used throughout the article. Fonts should not be changed from the default and neither should the text size of 10pt (with the exception being footnotesize, of course).

9.1 Parenthetical citations. In accordance with SIAM's editorial style manual [9], citations are allowed to be either parenthetical or not. For example, both "Note that several of the example are adapted from Mittlebach and Goossen's guide to LaTeX [7]." and "Note that several of the example are adapted from [7]." are both acceptable and such constructions are common.

9.2 Additional requested examples. An example of an en-dash in a number range, e.g., 1–9.

Since mentioning it is so common, use `\CC` to put a nicely formatted C++ in your document.

An example of `\textrm` within an equation to get non-math typesetting of some English words.

$$x = y - z \quad \text{for all } x.$$

Also note that `\operatorname` can be used to define a function like min, max, argmin, etc. (see Algorithm 7.1) so that it is formatted as roman text no matter where it appears.

10 Conclusion. A conclusion or summary is numbered like any other section.

A An example appendix. Appendices are created with the normal sectioning commands, following the command `\appendix`.

LEMMA A.1. *Test Lemma.*

B An appendix regarding the Bibliography. The SIAM BibTeX style file (called `siamplain.bst`), in addition to the standard key fields, includes the keys listed below:

- `doi`: Digital object identifier, a unique alphanumeric string
- `url`: Web address, usually impermanent
- `urldate`: Date that the web address was last accessed
- `eprint`: Archive identifier, a unique alphanumeric string

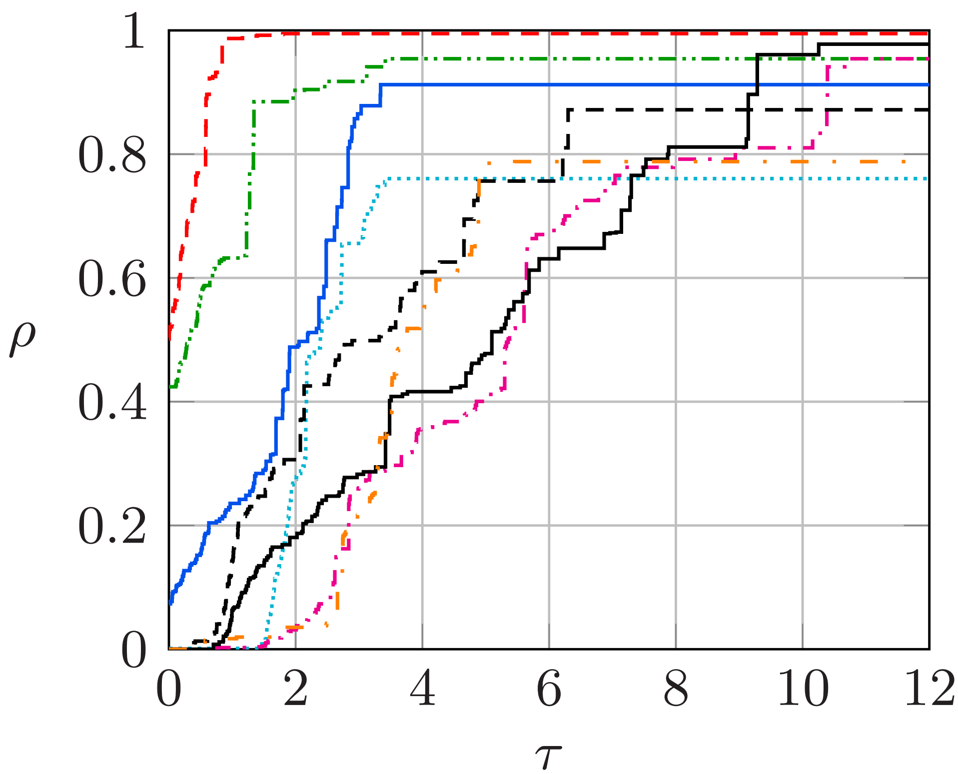

Figure 5.2: Example figure using an external image file that spans both columns.

- `eprintclass`: Archive class
- `archive`: Archive URL, defaults to https://arXiv.org/abs
- `archivepreprint`: Archive name, defaults to "arXiv"
- `eid`: Article ID, if there are no page numbers
- `pagetotal`: Total number of pages, for use with article ID

Every entry type includes an optional link to a DOI, a URL, and/or an archive preprint reference. Further, the `article` entry supports an Article ID, `eid`, and number of pages, `pagetotal`.

B.1 Additional bibliography examples. Here we provide additional examples for technical reports from Cornell [3] and the University of Zurich [8] as well as a conference paper from the Proceedings of the Seventeenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA) [1].

Acknowledgments. The acknowledgments section comes immediately before the references and after any appendices. It should be declared by `\section*{Acknowledgments}` so that it is not numbered.

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References

[1] G. Aggarwal and J. D. Hartline, *Knapsack auctions*, in Proceedings of the Seventeenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), Philadelphia, 2006, SIAM, pp. 1083–1092, http://dl.acm.org/citation.cfm?id=1109557.1109677.

[2] American Mathematical Society, *User's guide for the `amsmath` package (version 2.0)*, 2002, ftp://ftp.ams.org/pub/tex/doc/amsmath/amsldoc.pdf (accessed 2015-07-30).

[3] L. P. Chew, *Guaranteed-quality triangular meshes*, Technical Report 89-983, Department of Computer Science, Cornell University, Apr. 1989. https://hdl.handle.net/1813/6899.

[4] P. Dawkins, *Paul's online math notes: Calculus I — notes*, http://tutorial.math.lamar.edu/Classes/CalcI/MeanValueTheorem.aspx (accessed 2015-07-08).

[5] M. Downes, *Short math guide for LaTeX*, 2002, ftp://ftp.ams.org/pub/tex/doc/amsmath/short-math-guide.pdf (accessed 2015-07-30).

[6] G. H. Golub and C. F. Van Loan, *Matrix Computations*, The Johns Hopkins University Press, Baltimore, 4th ed., 2013.

[7] F. Mittlebach and M. Goossens, *The LaTeX Companion*, Addison-Wesley, 2nd ed., 2004.

[8] B. Stiller, T. Bocek, F. Hecht, G. Machado, P. Racz, and M. Waldburger, *Mobile systems IV*, tech. report, University of Zurich, Department of Informatics, Jan. 2010.

[9] *SIAM style manual: For journals and books*, 2013, https://epubs.siam.org/pb-assets/files/SIAM_STYLE_GUIDE_2019-1635349464967.pdf.